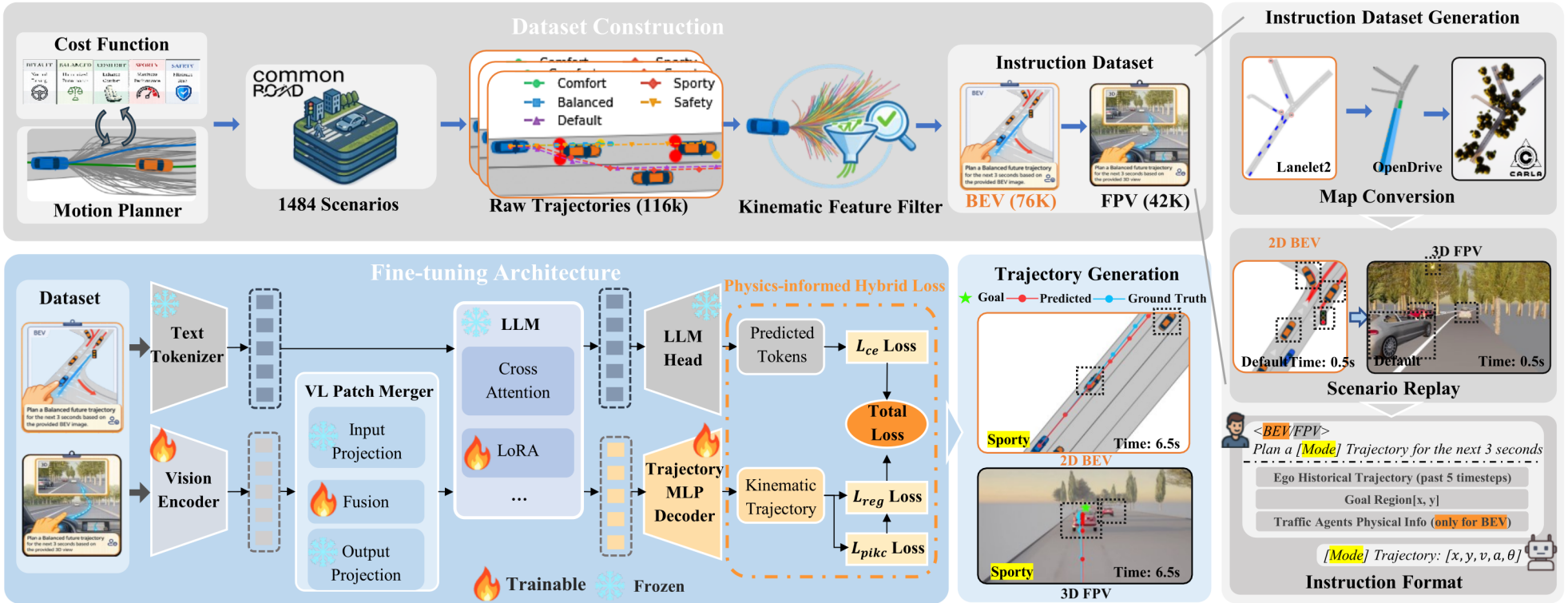
Dataset Construction
Cost Function
Motion Planner
common ROAD
1484 Scenarios
Comfort
Sporty
Balanced
Safety
Default
Raw Trajectories (116k)
Kinematic Feature Filter
Instruction Dataset
BEV (76K)
FPV (42K)
Instruction Dataset Generation
Lanelet2
OpenDrive
CARLA
Map Conversion
2D BEV
3D FPV
Default Time: 0.5s
Default Time: 0.5s
Scenario Replay
<BEV/FPV>
Plan a [Mode] Trajectory for the next 3 seconds
Ego Historical Trajectory (past 5 timesteps)
Goal Region[x, y]
Traffic Agents Physical Info (only for BEV)
[Mode] Trajectory: [x, y, v, a, θ]
Instruction Format
Fine-tuning Architecture
Dataset
Text Tokenizer
Vision Encoder
VL Patch Merger
Input Projection
Fusion
Output Projection
LLM
Cross Attention
LoRA
...
LLM Head
Trajectory MLP Decoder
Physics-informed Hybrid Loss
Predicted Tokens
Kinematic Trajectory
L_{ce} Loss
L_{reg} Loss
L_{pikc} Loss
Total Loss
Trainable
Frozen
Trajectory Generation
Goal
Predicted
Ground Truth
Sporty
Time: 6.5s
2D BEV
Sporty
Time: 6.5s
3D FPV